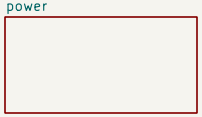
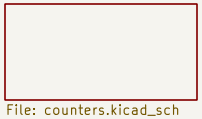
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counters



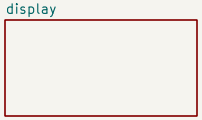
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hours



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display

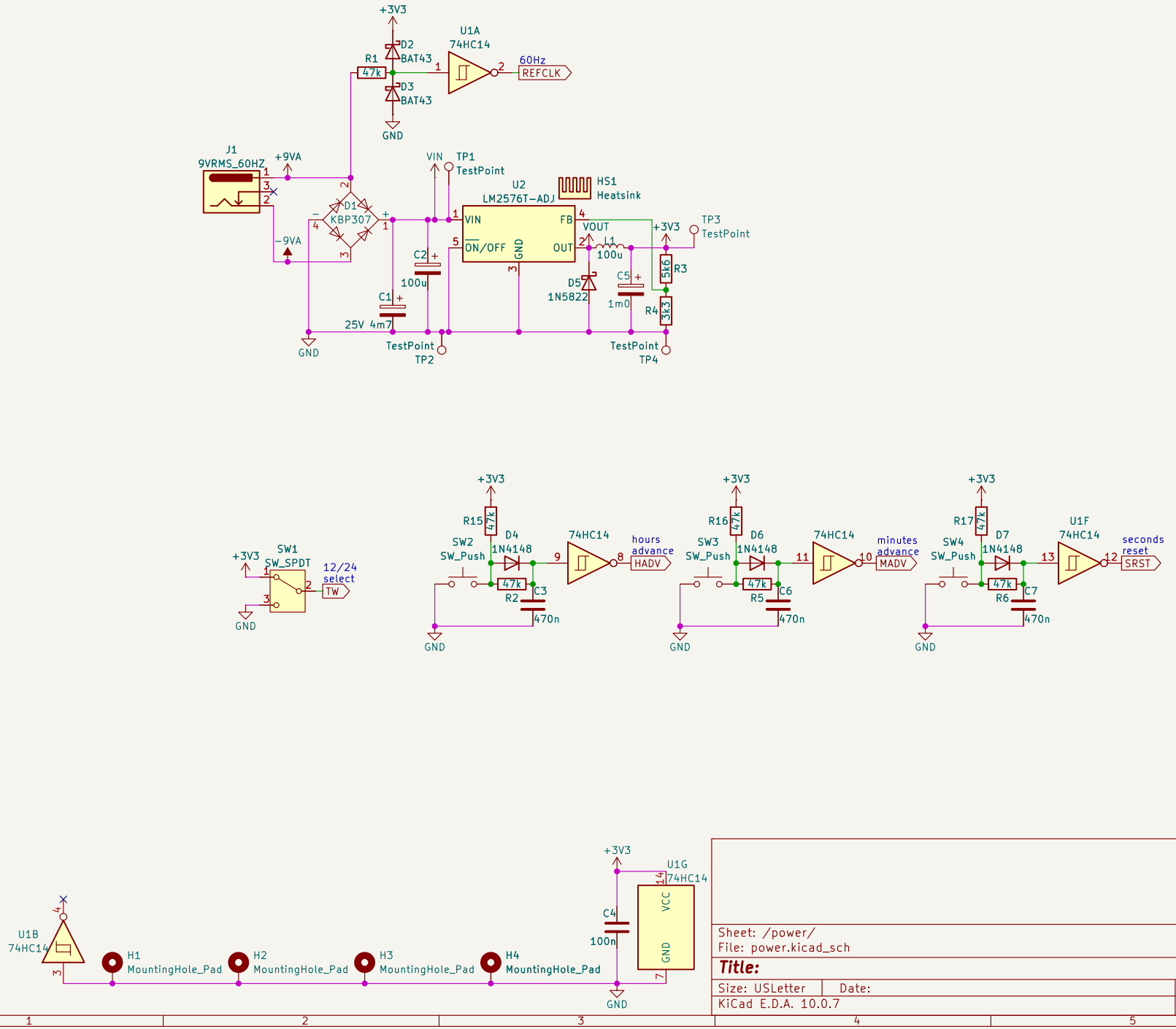


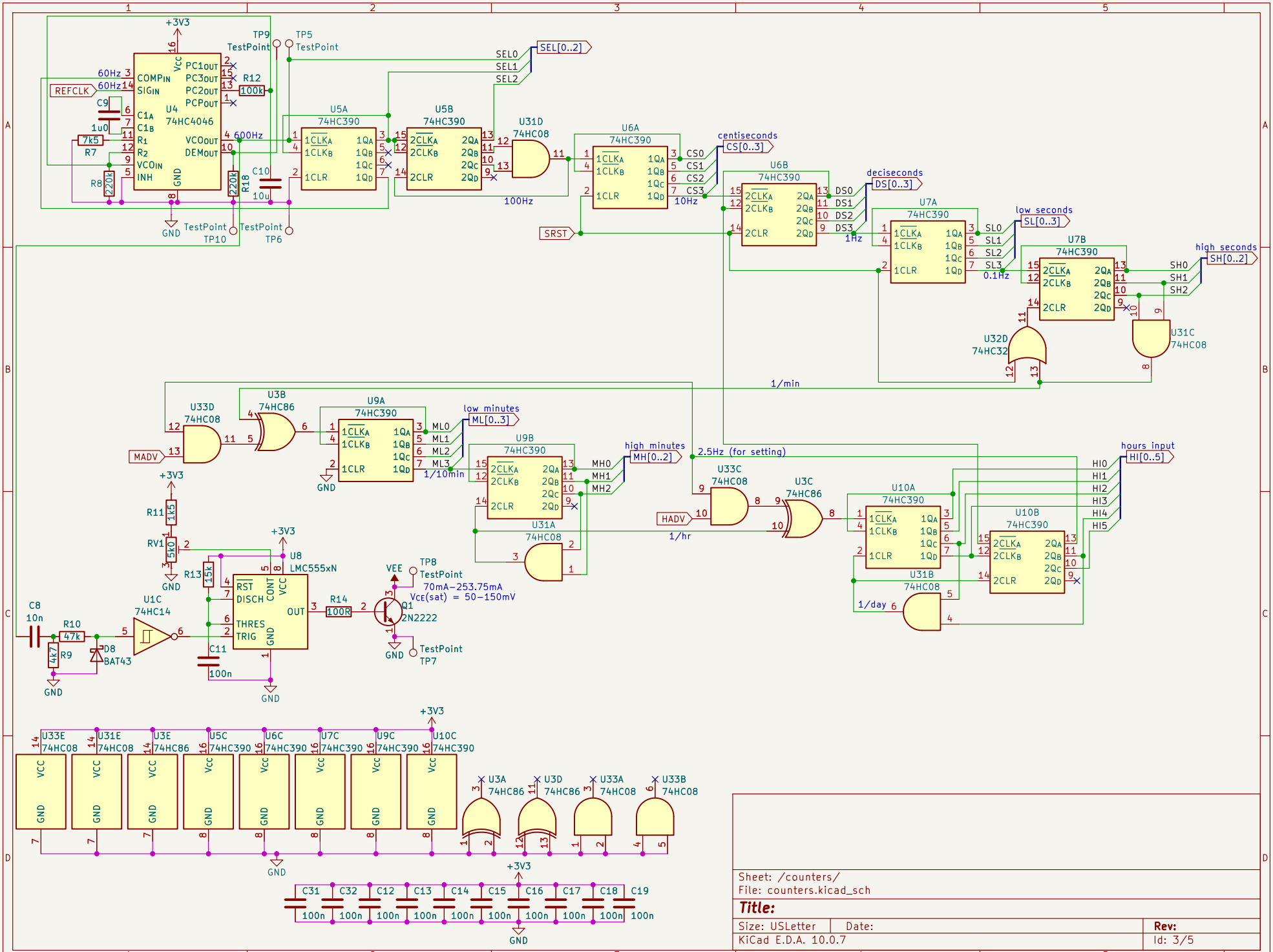
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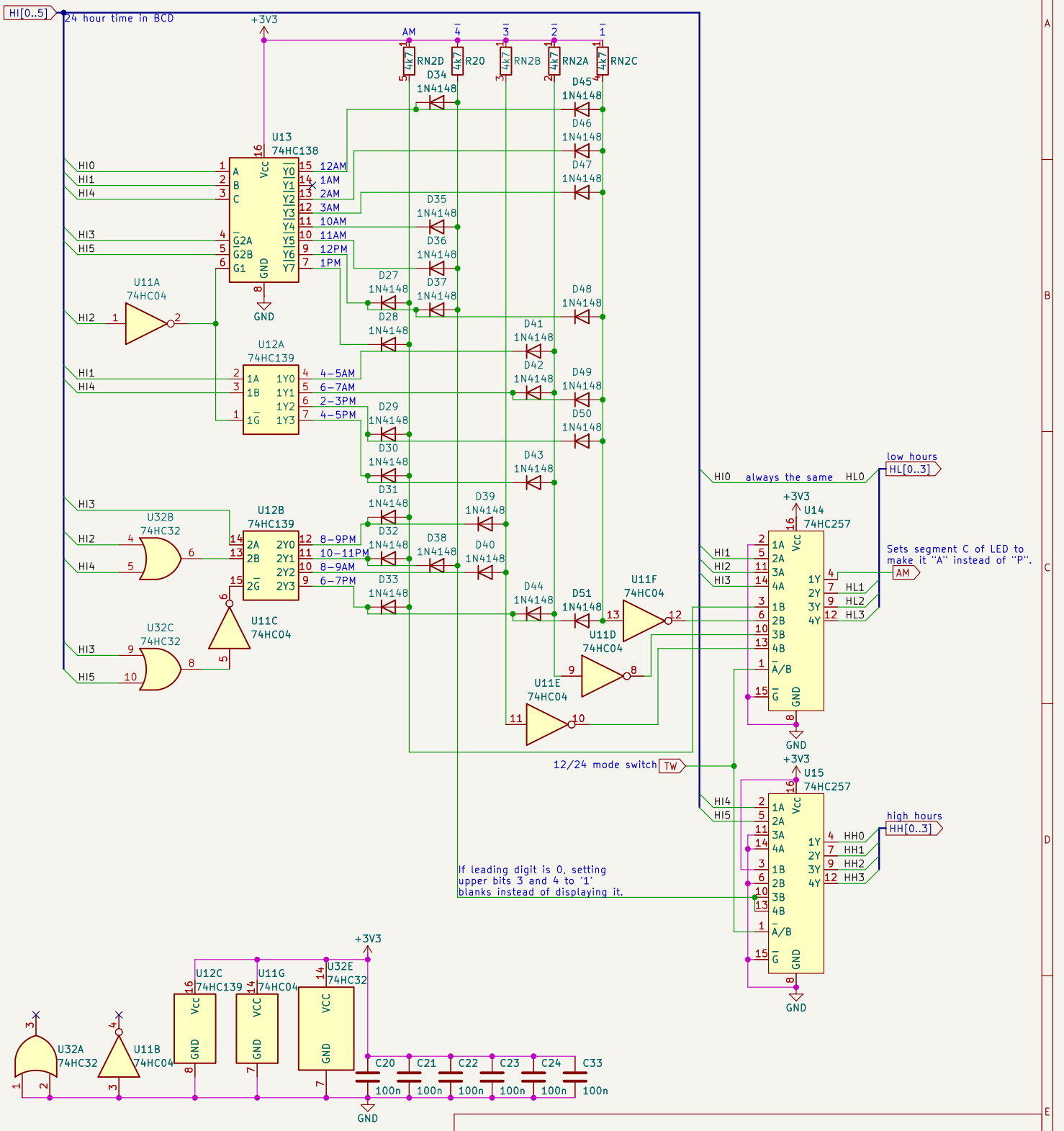
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**Title:**

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+3V3
U1A
74HC14
D2
BAT43
R1
47k
D3
BAT43
GND
60Hz
REFCLK
J1
9VRMS_60HZ
+9VA
−9VA
D1
KBP307
VIN
TP1
TestPoint
U2
LM2576T−ADJ
HS1
Heatsink
VIN
ON/OFF
GND
FB
OUT
VOUT
L1
100u
+3V3
TP3
TestPoint
R3
5k6
R4
3k3
C2
100u
C1
25V 4m7
D5
1N5822
C5
1m0
TestPoint
TP2
TestPoint
TP4
SW1
SW_SPDT
12/24
select
TW
R15
47k
SW2
SW_Push
D4
1N4148
74HC14
hours
advance
HADV
R2
47k
C3
470n
R16
47k
SW3
SW_Push
D6
1N4148
74HC14
minutes
advance
MADV
R5
47k
C6
470n
R17
47k
SW4
SW_Push
D7
1N4148
U1F
74HC14
seconds
reset
SRST
R6
47k
C7
470n
U1B
74HC14
H1
MountingHole_Pad
H2
MountingHole_Pad
H3
MountingHole_Pad
H4
MountingHole_Pad
U1G
74HC14
VCC
GND
C4
100n
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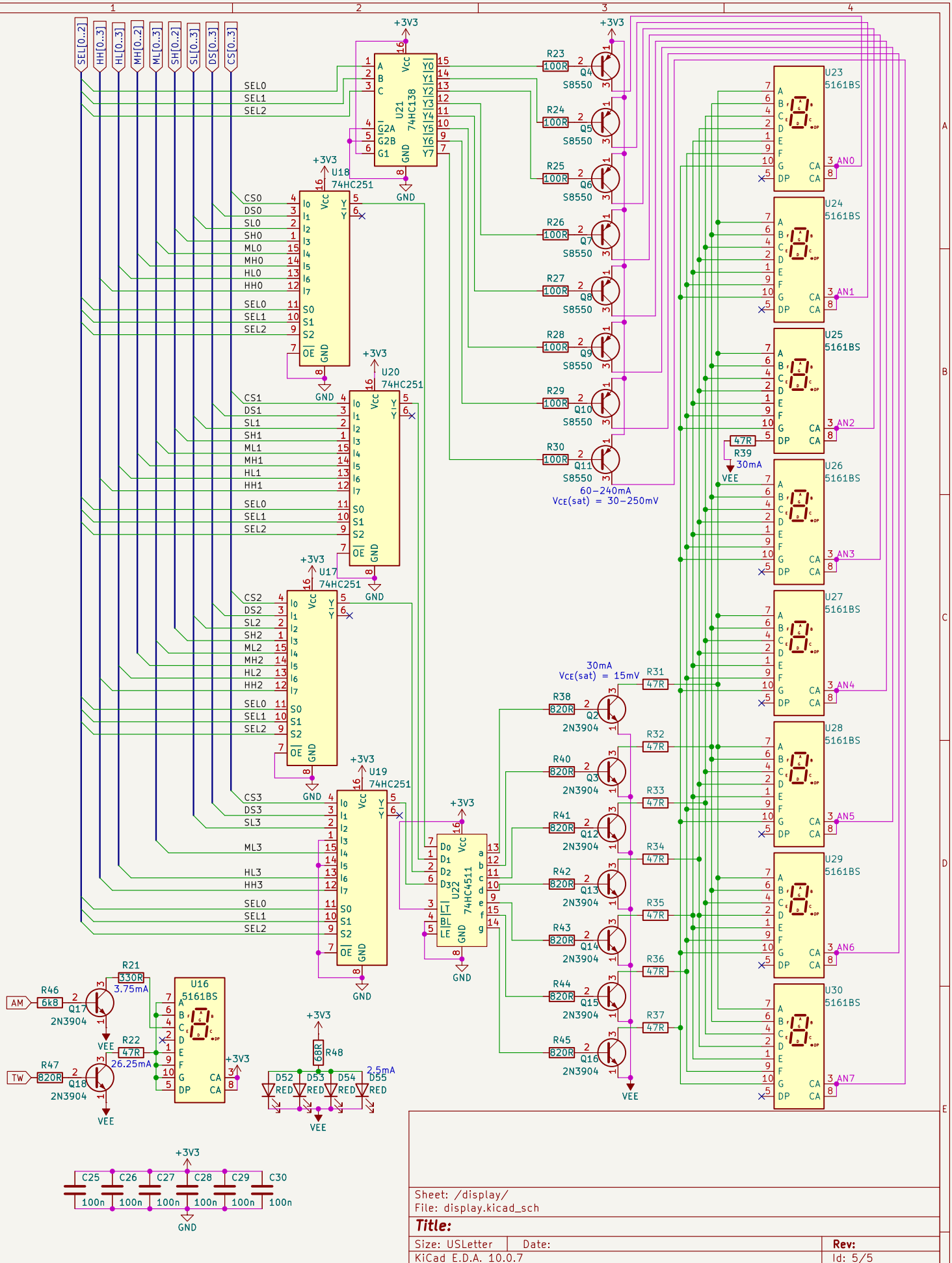

+3V3
TP9
TestPoint
TP5
TestPoint
SEL0
SEL1
SEL2
SEL[0..2]
60Hz
REFCLK
U4
74HC4046
COMPIN
SIGIN
PC1OUT
PC3OUT
PC2OUT
PCPOUT
VCOOUT
DEMOUT
VCOIN
INH
R12
100k
C9
1u0
R7
7k5
R8
220k
R18
220k
C10
10u
600Hz
GND
TestPoint
TP10
TestPoint
TP6
U5A
74HC390
U5B
74HC390
U31D
74HC08
100Hz
U6A
74HC390
centiseconds
CS[0..3]
CS0
CS1
CS2
CS3
10Hz
U6B
74HC390
deciseconds
DS[0..3]
DS0
DS1
DS2
DS3
1Hz
U7A
74HC390
low seconds
SL[0..3]
SL0
SL1
SL2
SL3
0.1Hz
U7B
74HC390
high seconds
SH[0..2]
SH0
SH1
SH2
SRST
U32D
74HC32
U31C
74HC08
1/min
U33D
74HC08
MADV
U3B
74HC86
U9A
74HC390
low minutes
ML[0..3]
ML0
ML1
ML2
ML3
1/10min
U9B
74HC390
high minutes
MH[0..2]
MH0
MH1
MH2
2.5Hz (for setting)
U33C
74HC08
HADV
U3C
74HC86
U31A
74HC08
1/hr
U10A
74HC390
U10B
74HC390
hours input
HI[0..5]
HI0
HI1
HI2
HI3
HI4
HI5
U31B
74HC08
1/day
+3V3
R11
1k5
RV1
5k0
GND
U8
LMC555xN
R13
15k
RST
DISCH
CONT
VCC
THRES
TRIG
OUT
GND
C11
100n
R14
100R
VEE
TP8
TestPoint
70mA–253.75mA
VCE(sat) = 50–150mV
Q1
2N2222
GND
TestPoint
TP7
C8
10n
R10
47k
R9
4k7
D8
BAT43
U1C
74HC14
U33E
74HC08
U31E
74HC08
U3E
74HC86
U5C
74HC390
U6C
74HC390
U7C
74HC390
U9C
74HC390
U10C
74HC390
VCC
GND
U3A
74HC86
U3D
74HC86
U33A
74HC08
U33B
74HC08
C31
C32
C12
C13
C14
C15
C16
C17
C18
C19
100n
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